

Soubhik Ghosh

Artificial Intelligence & Machine Learning Engineer

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Professional Summary

Full Stack & AI/ML Engineer with **4 years** of full-time experience (plus a 1-year internship) delivering scalable FinTech platforms at HDFC Bank, PayU, and FICO. Ships production Generative AI (LLM agents, RAG, document intelligence) and classical ML/Computer Vision (face/voice biometrics, anti-spoofing, signature verification) on robust full-stack architectures. Recognized with the **Silver Star Award** (HDFC) for presenting transformative AI to the CEO & Board, and the **Quarterly Ace Award** (PayU) for zero-defect delivery.

Technical Skills

Languages	Python, Java, TypeScript, JavaScript (ES6+), C/C++
GenAI / LLM	Gemini (1.5/2.5 Flash), Vertex AI, LiteLLM, LangChain, LangGraph, RAG, Agentic Frameworks, Prompt Engineering
ML / CV / NLP	PyTorch, Transformers, BERT, RoBERTa, DeBERTa, LayoutLM, Donut, YOLOv8, Pix2Pix GAN, ArcFace, FaceNet, InsightFace, Wav2Vec2, Whisper, Indic Parler-TTS, ECAPA-TDNN
Full Stack	ReactJS, Angular, Micro Frontends, ThreeJS, FastAPI, Spring Boot, Node.js, REST, Microservices
Data / Infra	PostgreSQL, MySQL, Couchbase, Redis, FAISS, BM25, Vector DB, YugabyteDB; Docker, Kubernetes, CI/CD, GCP, AWS, Kafka; Keycloak, FIDO2, OAuth, OWASP LLM Top 10

Experience

Data Scientist (Deputy Manager)

HDFC Bank

Nov 2024 - Present

Bengaluru, India

Generative Artificial Intelligence & Large Language Model Engineering

AI Fabric: TradeOps

LLM Document Intelligence Platform

Stack: Python, FastAPI, Gemini 2.5 Flash, Vertex AI, LiteLLM, LayoutLM, BERT, Docker, CI/CD

- Architected and led the bank's **first production Python AI platform**: document classification/extraction microservices for **800+ users pan-India** at **97% accuracy**, cutting manual processing time **90%**.
- Built a **pluggable LLM interface** (Vertex AI + LiteLLM) eliminating vendor lock-in, pairing Gemini with **LayoutLM/BERT** extraction, confidence scoring, a prompt hub, and an 11-API feedback loop (**98%+ test coverage**); led 4 engineers and set up CI/CD.

Cheque Processing: Clearing House

High-Throughput Document AI

Stack: Python, FastAPI, Gemini 1.5/2.5 Flash, Vertex AI, LiteLLM, LayoutLM, Donut, OpenCV

- Solely owned an AI cheque engine pairing **Gemini** with **LayoutLM/Donut**, hitting **99%+ accuracy** at **10+ cheques/sec**; built a validation harness over **700K+ ICR reject records**, projected to halve operations time.

VerifyX: Audit, Credit & Retail AI

RAG-Powered Verification Framework

Stack: ReactJS, FastAPI, Gemini, RAG (CAG agent), LiteLLM

- Built an extensible audit framework (VKYC + ECCS cheque-data audit) with a **CAG chat agent over RAG** spanning **27,000+ documents**, reused across retail and credit from one modular codebase.
- Optimized chunking and token strategy to cut LLM cost; **demoed live to the CEO and Board of Director**.

Narad AI & VRM: Virtual Relationship Manager

Enterprise Agentic Frontends

Stack: ReactJS, Micro Frontends (MFE), Gemini, FastAPI, LiteLLM, ThreeJS

- Single-handedly built the **Narad AI** email-agent frontend on a reusable MFE architecture (**98%+ test coverage**) and the **VERA** voice-to-voice interface with real-time **ThreeJS** 3D visualizations.

AI Banking Platform

LangGraph Agentic Orchestration

Stack: Python, LangGraph, FastAPI, Gemini, LiteLLM, ReactJS MFE, Keycloak/MFA

- Co-designed and drove a **LangGraph-based, skills-oriented agentic platform** composing banking workflows from modular skills, integrating **session management**, **MFA authentication**, and downstream banking entities behind a unified orchestration layer.
- Delivered the customizable **Aqua AI** micro-frontend chat surface and enforced strict modularity and code-quality standards across the platform's repositories.

RAG-as-a-Service

Shared Retrieval Infrastructure

Stack: Python, FAISS, BM25 (hybrid retrieval), Sentence-Transformers, Cohere/BGE reranker, LiteLLM

- Pioneered a hybrid **FAISS + BM25** retrieval service with sentence-transformer embeddings and a **Cohere/BGE reranker**, reducing LLM hallucinations by **40-60%** across 5 projects.

Machine Learning, Computer Vision & Biometrics

Signature Verification Pipeline

Document Forensics (CV)

Stack: YOLOv8, Pix2Pix GAN, Embedding Matching, Bicubic Interpolation, PyTorch

- Designed an end-to-end pipeline: **custom YOLOv8** for signature extraction, **Pix2Pix GAN** for image denoising/cleaning, and embedding-based similarity with bicubic interpolation for precise verification.

Pay-by-Face & LOOK: Facial Biometrics

Real-Time Recognition + Anti-Spoofing

Stack: ArcFace, FaceNet, InsightFace, FAISS/Milvus vector store, Custom Anti-Spoofing CNN, PyTorch

- Benchmarked **7 face-embedding models** (ArcFace, FaceNet variants on InsightFace) and shipped **sub-100ms** recognition at **99.2% accuracy** under production load, served from a FAISS/Milvus vector store.
- Built a **custom anti-spoofing CNN** using liveness, depth maps, and texture analysis, achieving a **false acceptance rate below 0.1%**; recognized with the **Silver Star Award**.

Vaani & Pay-by-Voice: Voice Biometrics

Speaker Verification + Voice-to-Voice

Stack: Indic Parler-TTS, Wav2Vec2, Whisper (STT), ECAPA-TDNN/x-vector speaker verification, PyTorch

- Fine-tuned **Indic Parler-TTS** and **Wav2Vec2/Whisper** STT and built **ECAPA-TDNN** speaker verification reaching **97% accuracy**, plus anti-spoofing voice forensics for a multimodal Pay-by-Voice system.

Ankan: Image Labelling Platform

Active-Learning Annotation Tool

Stack: ReactJS, FastAPI, YugabyteDB, Active Learning, Quality Gating

- Built from scratch an annotation tool with active-learning loops and quality gates; **100+ users** have labelled **10,000+ images**, now the standard for bank-wide ML training data.

Pareekshana Automation Suite & AI OS

QA Automation + Platform Foundation

Stack: ReactJS MFE, RAG-based CAG, FastAPI, LLM Orchestration

- Redesigned the enterprise QA suite (User Experience overhaul, Micro Frontend, RAG-based CAG) and laid the groundwork for AI OS; led up to **10 engineers** and presented strategy to the board. Built a YOLO-based merchant-verification POC and an LLM-powered merchant-onboarding compliance solution.

Software Engineer / Associate Software Engineer

Aug 2022 - Nov 2024

PayU (Wibmo)

Bengaluru, India

- Owned the **Angular frontend end-to-end** for the **Fraud Detection Risk Management Portal** (Angular, Figma), from design and development through state management, performance tuning, and production deployment of complex data-visualization dashboards.
- Architected and scaled **Spring Boot** microservices to absorb traffic spikes, cutting response times **50%**; developed **20+ REST APIs** backed by **Couchbase** and **Redis** caching
- Built the **FIDO2 / FIDO UAM / Keycloak** authentication and secure API gateways.
- Containerized and shipped services with **Docker, Kubernetes, and Nginx**; earned the **PayU ThankU and Quarterly Ace** awards for consistent full-stack delivery.

Associate Software Engineer / Software Engineering Intern

Jun 2021 - Aug 2022

Fair Isaac Corporation (FICO)

Bengaluru, India

- Built Python + Computer Vision pipelines to extract data from bank statements and identity documents; optimized C++/Python facial-recognition code toward **FRVT NIST certification** and annotated rPPG liveness-detection datasets within an Agile SDLC.

Open-Source Projects

soweak: AI Security Framework (Python & TypeScript)

github.com/SoubhikGhosh/soweak | PyPI & npm

Stack: Python, TypeScript, Transformers, RoBERTa, DeBERTa, NER, Sentence-Transformers, OWASP LLM Top 10

- Published an OWASP-aligned security **middleware that defends every boundary of an LLM pipeline** (input, retrieval, tool calls, output, streaming) with block/redact/transform/approval decisions and full audit trails; shipped both a Python (PyPI) library and an isomorphic TypeScript (npm) port for Node, browsers, and edge runtimes.
- Trained **custom NER and NLP ML classifiers** (RoBERTa/DeBERTa fine-tunes) for prompt-injection, jailbreak, PII/DLP, and toxicity detection, with LangChain, OpenAI, and Gemini adapters plus a red-team CLI.

aakaar: NL to DAG Workflow Automation

github.com/SoubhikGhosh/aakaar

Stack: Python, FastAPI, React, TypeScript, LLM Planner, WebSockets, SQLite, Chroma, Playwright

- Built a multi-tenant platform that compiles plain-language tasks into a typed **DAG of registry-defined capabilities** via an LLM planner, then executes it on a generic runtime interpreter with a credential vault and per-task audit.
- Engineered a remote-execution spine dispatching capability nodes over outbound WebSockets to lightweight cross-OS agents (shell, system, desktop-GUI), letting one authored workflow run on the server or any enrolled workstation.

Education

B.Tech, Computer Science and Engineering, KIIT, Bhubaneswar, **9.56 CGPA**

2018 to 2022

Class XII (CBSE), Delhi Public School, Dhanbad, **90.4%**

2018

Class X (ICSE), De-Nobili School CMRI, Dhanbad, **95%**

2016

Awards & Certifications

- Silver Star Award** (HDFC Bank), AI technical excellence, presented to CEO & Board; **Quarterly Ace & ThankU** Awards (PayU); delegate, **Google I/O Connect Bengaluru 2025** (HDFC Bank).
- Google IT Support Specialization; From Data to Insights with Google Cloud; Human-Centered Design; **JLPT N4** (Japanese Language Proficiency Test).